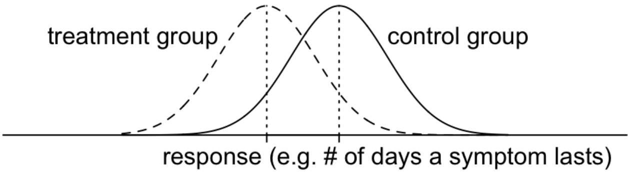
STAT 20000 Chapter 4
The Average and Standard Deviation (SD)

Making Comparisons using Histrograms

- Suppose some study turns out with histograms like:



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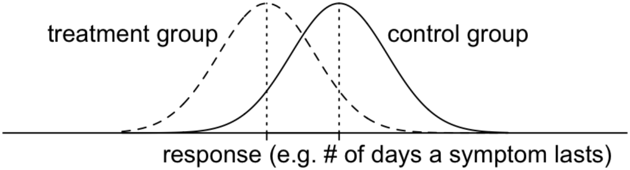
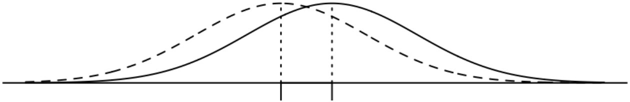


It would be natural to measure the effect of the treatment by the change in **center**.

- How are the results of this next study similar to those of the previous one? How are they different?



The greater the **spread** of the distributions, the less important is a given change in location.

Measures of Center & Spread

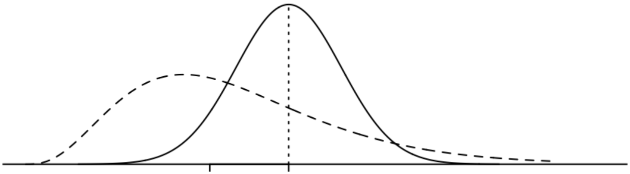
Center:

- Average (= Mean): the average value of the data
- Median: the middle value of the data
- Mode: the most frequent value of the data

Spread:

- Standard deviation (SD)
- Interquartile range (IQR) — See Ch5
- Range: maximum − minimum

Measures of **center** and **spread** are important summaries of a distribution. But watch out for situations where **they don't tell the whole story**:



- Always check the histograms. . .

Average

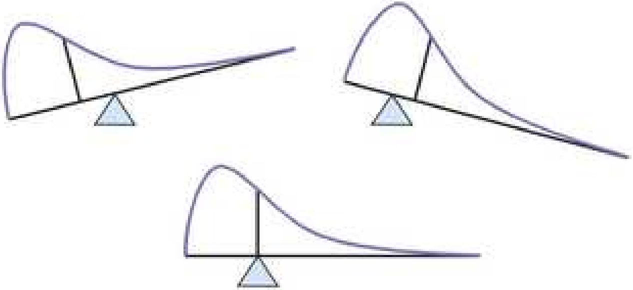
- The average of a list of numbers is simply

$$\text{Average} = \frac{\text{Sum of the values}}{\text{How many there are}}$$

- What does the average have to do with the center of the histogram?

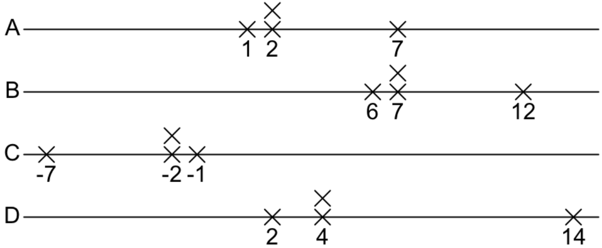
The Average and the Histogram

The average of a histogram is the balance point of the histogram, if it were a solid mass.



Effects of Shifting, Reflection, and Scaling on the Average

Consider these four lists:



1. How is list B related to list A? How is the average of list B related to the average of list A?

2. Ditto, for C?

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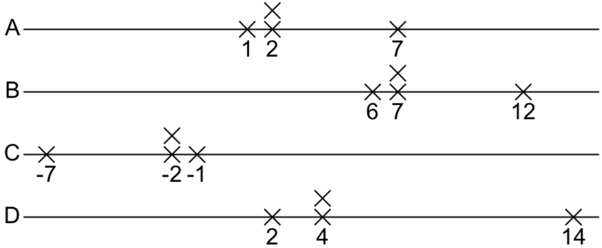

1. How is list B related to list A? How is the average of list B related to the average of list A?
   The numbers shifted by 5, and so does their average.

2. Ditto, for C?

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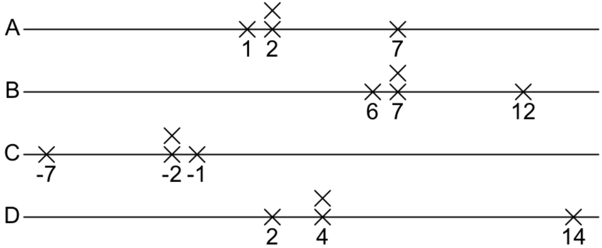

1. How is list B related to list A? How is the average of list B related to the average of list A?
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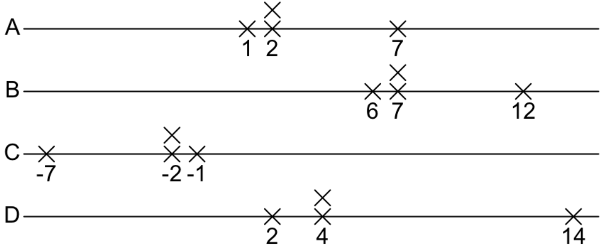
2. Ditto, for C?
   The numbers changed sign, and so did their average.



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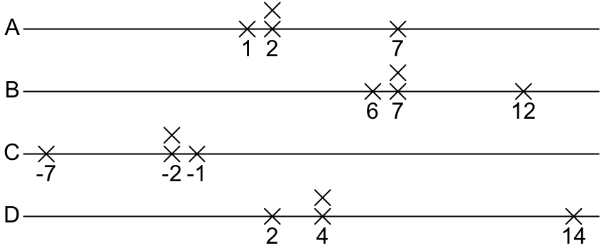
General Rules:

- If each element of a list is shifted by some constant, the average will be **shifted** by that constant.
- If each element of a list is multiplied by some constant, the average will be **multiplied** by that constant.



3. Ditto, for D?
   The numbers doubled, and so did their average.

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- If each element of a list is shifted by some constant, the average will be **shifted** by that constant.
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Another Measure of the Center: Median

The *median* of a numerical variable is a number such that half of the observed value are smaller than it and half are larger than it.

Ex 1: Suppose a variable has 5 observed values: 4, 8, 3, 5, 12.

$$\begin{array}{rcccccc} \text{data} & \rightarrow & 4 & 8 & 3 & 5 & 12 \\ \text{sorted} & \rightarrow & 3 & 4 & 5 & 8 & 12 \end{array}$$

Ex 2: If the variable has one more observation: 4, 8, 3, 5, 12, *100*.

$$\begin{array}{rccccccc} \text{data} & \rightarrow & 4 & 8 & 3 & 5 & 12 & 100 \\ \text{sorted} & \rightarrow & 3 & 4 & 5 & 8 & 12 & 100 \end{array}$$

The median is thus $\dfrac{5+8}{2} = 6.5$.

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The median is thus $\frac{5+8}{2} = 6.5$.

Median is More Robust to Extreme Values Than Average

Example. If the variable has a new 6th observation of value 100,

$$4,\ 8,\ 3,\ 5,\ 12,\ \mathit{100}$$

The new average of the variable becomes

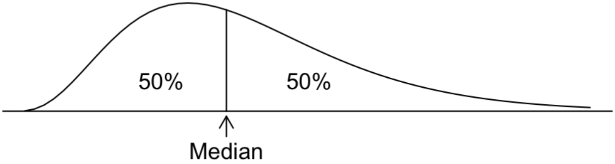
$$\frac{4 + 8 + 3 + 5 + 13 + 100}{6} = \frac{132}{6} = 22.$$

Data	Mean	Median
4, 8, 3, 5, 12	6.4	5
4, 8, 3, 5, 12, 100	22	6.5

The median is less affected by the extreme value (100) than the average. We say the median is more *robust* than the mean.

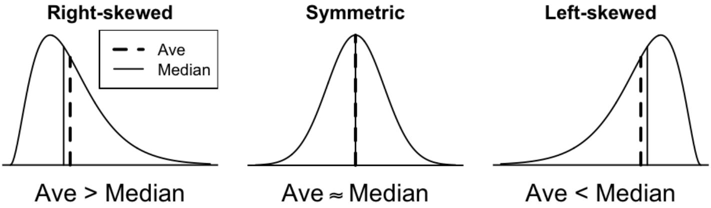
Median and Average of a Histogram

The **median** divides the area of a histogram evenly.



Average vs. Median and Skewness

- In a symmetric distribution, Average $\approx$ Median.
  - If exactly symmetric, then Average $=$ Median.
- In a skewed distribution, the average is pulled toward the longer tail.



Measuring Spread: Standard Deviation (SD)

Steps to compute the SD of the list 1,2, 2,7

1. Find the average of the list.
2. Find the deviations from average.
3. Square the deviations.
4. Average the squared deviations.
5. Take the square root.

SD = Root Mean Square (**RMS**) of the deviations from the average of the list

Measuring Spread: Standard Deviation (SD)

Steps to compute the SD of the list 1,2, 2,7

1. Find the average of the list. $\frac{1+2+2+7}{4} = 3$
2. Find the deviations from average.
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Steps to compute the SD of the list 1,2, 2,7

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5. Take the square root.

$$\text{SD} = \sqrt{22/4} \approx 2.3$$

SD = Root Mean Square (**RMS**) of the deviations from the average of the list

Example

Each of the following lists has an average of 50. Which list has the smallest SD? The largest?

A: 0, 20, 40, 50, 60, 80, ,100

B: 0, 48, 49, 50, 51, 52, ,100

C: 0, 1, 2, 50, ,98, 99,100

What are the SDs?

	A	B	C
By eye			
By calculator			

SD is very sensitive to extreme values.

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What are the SDs?

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By eye	middle	smallest	largest
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	A	B	C
By eye	middle	smallest	largest
By calculator	32	27	45

List B shows that the SD pays considerable attention to large deviations.

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Effects of Shifting, Reflection, and Scaling on the SD

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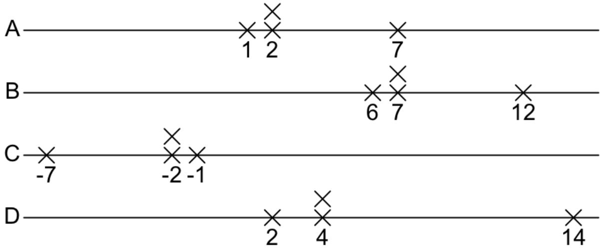

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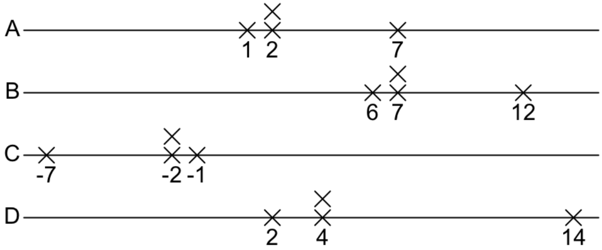

1. How is the SD of list B related to that of list A?
   The spread of a list is not affected by shifting.
   SD of B = SD of A .
2. Ditto, for C?

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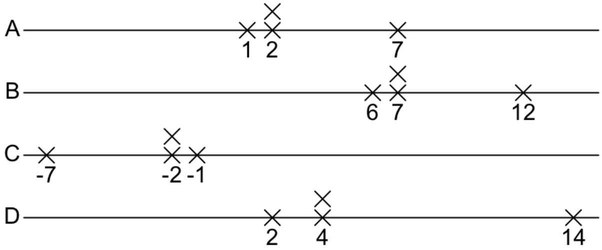

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   $$\text{SD of B} = \text{SD of A}$$
   .
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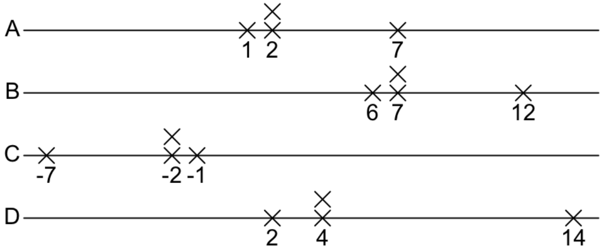
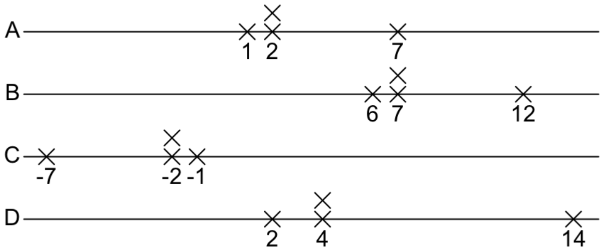
   The spread of a list is not affected by reflection.
   $$\text{SD of C} = \text{SD of A.}$$



3. Ditto, for D?

General Rule:

- If each element of a list is shifted by some constant, the SD is **unchanged**.
- If each element of a list is multiplied by some constant, the SD gets **multiplied** by the **absolute value** of that constant.



3. Ditto, for D? The spread doubled, and so did the SD.

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- If each element of a list is shifted by some constant, the SD is **unchanged**.
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Exercise

For list A, Average = 5, SD = 2.05.
Please find the averages and SDs for list B and list C respectively.

	List		
	A	B	C
	4	24	40
	6	26	60
	8	28	80
	6	26	60
	5	25	50
	1	21	10
	7	27	70
	6	26	60
	5	25	50
	2	22	20
Average	5.00		
SD	2.05		

The 68%- and 95%-Rules

Roughly 68% (2 in 3) of the entries on a list are within 1 SD of the average, the other 32% are further away.

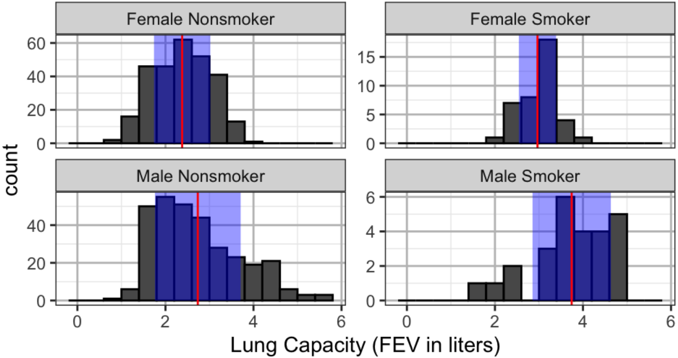
Roughly 95% (19 in 20) of the entries on a list are within 2 SDs of the average; the other 5% are further away.

This is so for many lists, but not all.

- These rules work especially well for lists with **bell-shaped** histograms.
- These rules work reasonably well for lists with unimodal, and not seriously skewed histograms.

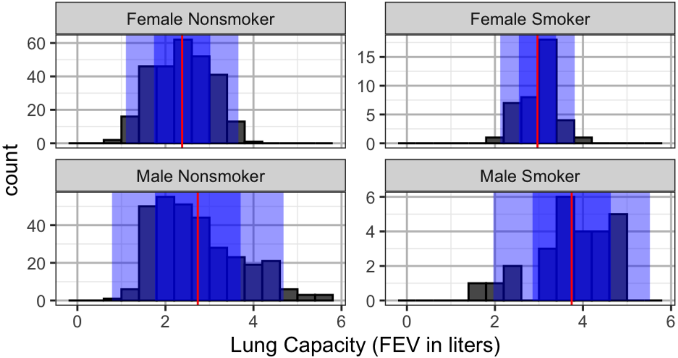
68% Rule for the FEV Data

Sex/Smoke	Average	SD	Count	proportion within *1 SD* from Ave
Female Nonsmoker	2.379	0.639	279	$170/279 \approx 60.9\%$
Female Smoker	2.966	0.423	39	$27/39 \approx 69.2\%$
Male Nonsmoker	2.734	0.974	310	$203/310 \approx 65.5\%$
Male Smoker	3.743	0.889	26	$17/26 \approx 65.4\%$



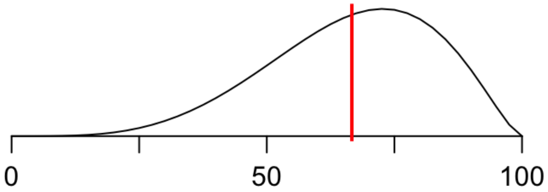
95% Rule for the FEV Data

Sex/Smoke	Average	SD	Count	proportion within *2 SDs* from Ave
Female Nonsmoker	2.379	0.639	279	$270/279 \approx 96.8\%$
Female Smoker	2.966	0.423	39	$38/39 \approx 97.4\%$
Male Nonsmoker	2.734	0.974	310	$298/310 \approx 96.1\%$
Male Smoker	3.743	0.889	26	$24/26 \approx 92.3\%$



Exercise

Is the SD of the histogram below closest to 5, 15, or 50?



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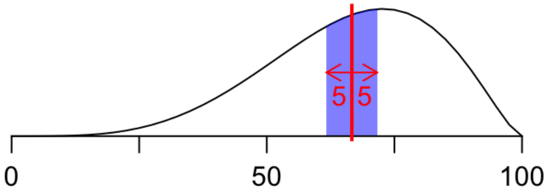

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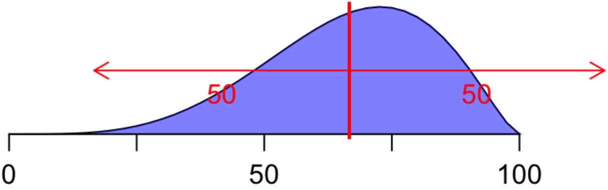

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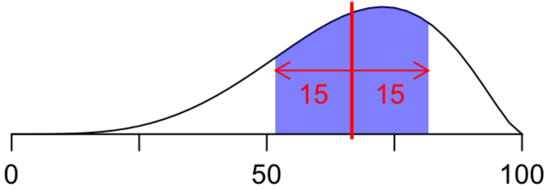
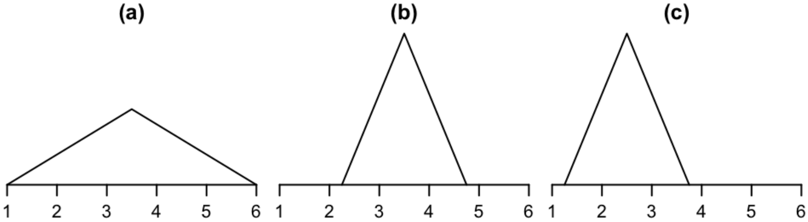


Exercise (68%- and 95%-Rules)

Below are sketches of histograms for three lists. Match the sketch with the description. Some descriptions will be left over. Give your reasoning in each case.

(i)	ave ≈ 3.5, SD ≈ 1	(iv)	ave ≈ 2.5, SD ≈ 1
(ii)	ave ≈ 3.5, SD ≈ 0.5	(v)	ave ≈ 2.5, SD ≈ 0.5
(iii)	ave ≈ 3.5, SD ≈ 2	(vi)	ave ≈ 4.5, SD ≈ 0.5



Mathematical Notation

- In mathematical notation, a generic list can represented as

$$x_1,\, x_2,\, \ldots, x_n.$$

  Here x_1 stands for the first element on the list, x_2 for the second, ..., and x_n for the last; n stands for the number of elements on the list.

- The average of the generic list is

$$\frac{\text{sum of the values}}{\text{how many there are}} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

  The average of $x_1, x_2, \ldots, x_n$ is commonly denoted by $\bar{x}$, read *x-bar*.

Caution: Sample SD

The SD of the generic list is the RMS deviation from average:

$$\text{SD} = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n}}$$

Most calculators and statistical softwares compute the SD as the slightly larger number

$$\text{sample SD} = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n-1}}$$

called the *sample SD*, commonly denoted by s.
Thus

$$\text{SD} = \sqrt{\frac{n-1}{n}}\, s = \sqrt{\frac{\text{number of entries} - 1}{\text{number of entries}}}\, (\text{sample SD}).$$

Summary of Measures of Center & Spread

	Shifting $x_i \to x_i + b$ for all i	Scaling $x_i \to ax_i$ for all i	Reflection $x_i \to -x_i$ for all i	Sensitive to extreme values?
Average $\bar{x}$	shifted by b $\bar{x} \to \bar{x} + b$	Scaled by a $\bar{x} \to a\bar{x}$	change sign $\bar{x} \to -\bar{x}$	Yes
Median	shifted by b	Scaled by a	change sign	No
SD	unchanged	Scaled by $\|a\|$ SD $\to \|a\| \times$ SD	unchanged	Yes